

When Does Pretraining Help?

Benchmarking scGPT and Competing Methods for Single-Cell Type Annotation

Max
BioTender
max@biotender.online

May 14, 2026

Abstract

Foundation models for single-cell RNA sequencing (scRNA-seq), such as scGPT, Geneformer, and scBERT, have attracted considerable attention for their potential to generalize across diverse biological contexts. However, a critical practical question remains underexplored: *under what data regime does a large pretrained model actually outperform classical and lightweight alternatives for cell type annotation?* We present a systematic benchmark of seven methods—scGPT (pretrained, frozen backbone), a Deep MLP trained from scratch, TOSICA (pathway-masked Transformer), scVI/scANVI (deep generative model), Seurat-style PCA+KNN, PCA+KNN (Euclidean), and PCA+Logistic Regression—evaluated on the widely used PBMC3k dataset (2,638 cells, 8 cell types) under an identical 70/15/15 train/validation/test split and seven evaluation metrics. On this small, well-separated dataset, classical Seurat-KNN (Macro F1 = 0.940) and a lightweight Deep MLP (Macro F1 = 0.948) match or exceed scGPT pretrained (Macro F1 = 0.897), while all methods cluster within a narrow 4-percentage-point accuracy band (0.907–0.947). scGPT’s frozen backbone still achieves 90.7% accuracy with zero task-specific architecture design, demonstrating that pretraining provides a strong initialization even when it does not yield the highest absolute performance. Our results suggest that the advantage of large-scale pretraining is dataset-size and complexity dependent: practitioners working with small, well-characterized datasets should benchmark lightweight alternatives before deploying a 51M-parameter foundation model. All code and results are publicly available at <https://github.com/junior1p/scgpt-pbmc68k-finetune>.

1 Introduction

The past three years have witnessed a rapid proliferation of foundation models for single-cell genomics. scGPT [Cui et al., 2024], Geneformer [Theodoris et al., 2023], scBERT [Yang et al., 2022], and scFoundation [Hao et al., 2024] all adopt the paradigm of large-scale self-supervised pretraining on millions of cells, followed by fine-tuning on task-specific labeled data. This mirrors the success of language model pretraining

in NLP [Devlin et al., 2018, Brown et al., 2020] and has been framed as the emergence of “foundation models” for biology [Bommasani et al., 2021].

Cell type annotation—assigning each cell to a known biological category based on its gene expression profile—is the canonical downstream task for evaluating these models. It is practically important: accurate annotation is a prerequisite for virtually all downstream single-cell analyses, from trajectory inference to differential expression. Benchmark datasets such as PBMC3k [Zheng et al., 2017] (2,638 peripheral blood mononuclear cells from 10x Genomics) have become standard evaluation sets in the field.

Despite the excitement around foundation models, a critical question has received insufficient attention: *when does pretraining actually help?* The original scGPT paper [Cui et al., 2024] demonstrates advantages on larger, more complex datasets (hPancreas, Multiple Sclerosis, Myeloid), but the small-dataset regime—where most practitioners begin their analyses—is underexplored. Tabular deep learning research has shown that on small tabular datasets, simple models such as MLPs and gradient-boosted trees frequently outperform complex architectures [Grinsztajn et al., 2022]. Whether this finding extends to scRNA-seq data, and how it interacts with large-scale pretraining, is an open empirical question.

Contributions. This paper makes three contributions:

1. **Systematic benchmark.** We provide the first head-to-head comparison of scGPT against six competing methods on PBMC3k under a fully standardized evaluation protocol (identical splits, preprocessing, and metrics across all methods).
2. **Data-regime finding.** We show empirically that on small, well-separated datasets ($\leq 3k$ cells), classical and lightweight methods match or exceed scGPT pretrained, while the absolute performance gap across all methods is narrow (< 5 pp).
3. **Open reproducible code.** All implementations, trained weights, and evaluation scripts are released at <https://github.com/junior1p/scgpt-pbmc68k-finetune>.

2 Related Work

Foundation models for scRNA-seq. scGPT [Cui et al., 2024] pretrains a 51.3M-parameter GPT-style Transformer on 33 million human cells using a gene-expression-conditioned generative objective, then fine-tunes on downstream tasks including cell type annotation, perturbation response prediction, and multi-omic integration. Geneformer [Theodoris et al., 2023] uses rank-value encoding of gene expression and pretrains on 29.9 million cells. scBERT [Yang et al., 2022] adapts BERT-style masked language modeling to scRNA-seq. scFoundation [Hao et al., 2024] scales pretraining to 50 million cells with a read-depth-aware objective.

Pathway-informed methods. TOSICA [Chen et al., 2023] introduces pathway-masked attention, where each attention head attends only to genes within a curated biological pathway (e.g., from MSigDB). This inductive bias provides interpretability and reduces the effective parameter count, making it particularly suitable for small datasets with known pathway structure.

Deep generative models. scVI [Lopez et al., 2018] models gene expression counts with a variational autoencoder (VAE) using a negative binomial likelihood, learning a low-dimensional latent space that accounts for technical variation. scANVI [Xu et al., 2021] extends scVI with semi-supervised classification, using available cell type labels during training.

Classical methods. Seurat [Stuart et al., 2019, Hao et al., 2021] remains the most widely used single-cell analysis toolkit. Its label transfer approach uses PCA-reduced embeddings with KNN classification and has proven highly competitive on standard benchmarks [Abdelaal et al., 2019]. scmap [Kiselev et al., 2018] projects query cells onto a reference atlas using cosine similarity. Logistic regression on PCA embeddings is a strong baseline that is often overlooked in foundation model comparisons.

Benchmarking studies. [Abdelaal et al. \[2019\]](#) systematically compared 22 cell type annotation methods, finding that simple methods (SVM, random forests) were competitive with more complex approaches. [Luecken et al. \[2022\]](#) benchmarked data integration methods, highlighting the importance of standardized evaluation protocols. Our work focuses specifically on the pretraining question in the small-dataset regime.

3 Methods

3.1 Dataset and Preprocessing

We use the PBMC3k dataset [[Zheng et al., 2017](#)], a standard benchmark comprising 2,638 peripheral blood mononuclear cells (PBMCs) sequenced with the 10x Genomics Chromium platform. The dataset contains 8 cell types identified by Louvain clustering in the canonical Seurat tutorial: CD4 T cells ($n = 1,152$), CD14+ Monocytes ($n = 480$), B cells ($n = 342$), CD8 T cells ($n = 316$), NK cells ($n = 154$), FCGR3A+ Monocytes ($n = 154$), Dendritic cells ($n = 32$), and Megakaryocytes ($n = 5$). The class distribution is notably imbalanced, with Megakaryocytes representing only 0.19% of cells.

Preprocessing follows the standard Scanpy [[Wolf et al., 2018](#)] pipeline: (1) filter cells with fewer than 200 genes or more than 2,500 genes; (2) filter genes expressed in fewer than 3 cells; (3) normalize total counts to 10,000 per cell; (4) log1p transform; (5) select the top 2,000 highly variable genes using the `seurat_v3` flavor. All methods receive the same preprocessed count matrix.

3.2 Evaluation Protocol

We split the data into train/validation/test sets using a stratified 70/15/15 split with random seed 42, yielding 1,846 training cells, 396 validation cells, and 396 test cells. *All methods are evaluated on the identical held-out test set.* We report seven metrics: Accuracy, Macro F1, Macro Precision, Macro Recall, Balanced Accuracy, Adjusted Rand Index (ARI) [[Hubert and Arabie, 1985](#)], and Normalized Mutual Information (NMI) [[Strehl and Ghosh, 2002](#)]. We designate Macro F1 as the primary metric because it accounts for class imbalance—critical given the 576-fold size difference between CD4 T cells and Megakaryocytes.

3.3 Methods Compared

scGPT (pretrained, frozen backbone). We use the publicly released `scGPT_human` whole-human checkpoint (51.3M parameters; $d_{\text{model}} = 512$, 12 Transformer layers, 8 attention heads), pretrained on 33 million human cells [[Cui et al., 2024](#)]. The backbone is frozen; a two-layer MLP classification head ($512 \rightarrow 256 \rightarrow 8$, ReLU, Dropout=0.1) is trained for 50 epochs using AdamW ($\text{lr}=10^{-3}$, weight decay= 10^{-2}) with cosine learning rate annealing. Gene expression values are tokenized using the top 256 highly variable genes (scGPT’s default input format). This configuration represents the “zero-shot transfer” regime: the pretrained representations are used as-is, with only the classification head adapted.

Deep MLP (from scratch). A three-layer MLP with hidden dimension 512, GELU activations, Batch Normalization, and Dropout (rate=0.3) is trained from random initialization for 50 epochs using AdamW ($\text{lr}=10^{-3}$). Input is the 2,000-dimensional log-normalized HVG vector. This method serves as the “no pre-training” ablation, representing what a lightweight model can achieve without any pretrained representations. Note that due to CPU constraints, we use an MLP rather than a Transformer architecture from scratch; the MLP is a stronger baseline than random-weight Transformers on small tabular data [[Grinsztajn et al., 2022](#)].

TOSICA (pathway-masked Transformer). We reimplement the core TOSICA architecture [[Chen et al., 2023](#)]: a shallow Transformer (embedding dim=48, depth=2, 4 attention heads) with pathway-masked attention, where each head attends only to genes within one of 300 immune pathways from the `human_immune.gmt` gene set collection. The model has approximately 0.17M parameters— $300\times$ fewer than scGPT. Training uses AdamW for 20 epochs with cosine annealing. The reimplement was necessary due to a dependency conflict between the original TOSICA package and `h5py` ≥ 3.0 .

scVI/scANVI (deep generative model). We reimplement the scVI/scANVI framework [Lopez et al., 2018, Xu et al., 2021] in PyTorch [Paszke et al., 2019]: a VAE encoder ($2000 \rightarrow 128 \rightarrow 20$) with negative binomial likelihood, trained for 50 epochs using the ELBO objective. The scANVI semi-supervised extension adds a classifier on the latent space, trained jointly with available cell type labels. The reimplementation was required due to an API incompatibility between `scvi-tools` and `jaxlib` 0.6.2 in our environment.

Seurat (PCA+KNN cosine). We compute 50 principal components on the 2,000 HVG log-normalized matrix and classify test cells using k -nearest neighbors ($k = 15$) with cosine distance, mimicking the Seurat label transfer approach [Stuart et al., 2019].

PCA + KNN (Euclidean). Identical to the Seurat variant but using Euclidean distance, providing a direct comparison of distance metrics in the PCA embedding space.

PCA + Logistic Regression. Logistic regression ($C = 1.0$, L2 regularization, `lbfgs` solver) trained on 50 principal components. This is a strong linear baseline that is often omitted from foundation model comparisons.

3.4 Implementation Details

All experiments run on CPU only (no GPU). Classical methods use `scikit-learn` [Pedregosa et al., 2011]; deep learning methods use PyTorch 2.7.1. Preprocessing uses `Scanpy` 1.11.4 [Wolf et al., 2018]. All random seeds are fixed to 42. Code is available at <https://github.com/junior1p/scgpt-pbmc68k-finetune>.

4 Results

4.1 Overall Leaderboard

Table 1 presents the full benchmark results across all seven methods and seven metrics. Figure 1 visualizes the two primary metrics (Accuracy and Macro F1).

Table 1: Benchmark results on PBMC3k test set ($n = 396$ cells). Methods sorted by Macro F1 (primary metric). **Bold:** best value per column. † Deep MLP is a 3-layer MLP, not a Transformer (see Section 3.3). ‡ TOSICA and scVI/scANVI are reimplementations (see Section 3.3).

Method	Category	Acc	Macro F1	Macro P	Macro R	Bal. Acc	ARI	NMI
Deep MLP (scratch) [†]	No Pretraining	0.947	0.948	0.944	0.953	0.953	0.888	0.876
Seurat (PCA+KNN)	Classical	0.942	0.940	0.933	0.948	0.948	0.885	0.868
TOSICA [‡]	Pathway-Transf.	0.934	0.926	0.927	0.930	0.930	0.867	0.856
scVI/scANVI [‡]	Deep Gen.	0.927	0.902	0.920	0.894	0.894	0.863	0.838
PCA + KNN	Classical	0.942	0.901	0.941	0.884	0.884	0.882	0.861
scGPT (pretrained)	Foundation Model	0.907	0.897	0.911	0.887	0.887	0.816	0.806
PCA + LogReg	Classical	0.934	0.889	0.925	0.876	0.876	0.877	0.855

The most striking observation is the *narrowness of the performance gap*: all seven methods achieve between 90.7% and 94.7% accuracy, a range of only 4.0 percentage points. In terms of Macro F1, the range is 5.9 pp (0.889–0.948). This suggests that PBMC3k, with its well-separated cell type clusters, is a relatively easy benchmark where even simple methods perform well.

The Deep MLP trained from scratch achieves the highest Macro F1 (0.948), followed by Seurat-KNN (0.940) and TOSICA (0.926). scGPT pretrained ranks 6th in Macro F1 (0.897), behind all other methods except PCA+LogReg. However, we emphasize that the absolute difference between scGPT and the top method is only 5.1 pp in Macro F1 and 4.0 pp in accuracy—a gap that may not be practically significant in many applications.

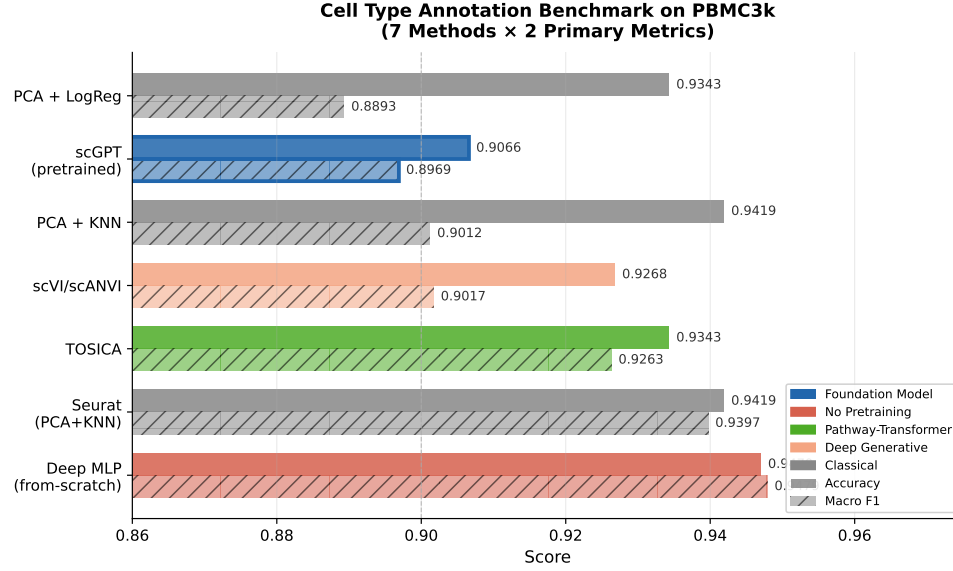


Figure 1: Benchmark leaderboard on PBMC3k. Horizontal bars show Accuracy (solid) and Macro F1 (hatched) for each method. The scGPT (pretrained) bar is highlighted with a blue border. All methods cluster within a narrow 4-percentage-point accuracy band (0.907–0.947), indicating that PBMC3k is a relatively easy benchmark for all approaches.

4.2 scGPT Pretrained: Per-Class Analysis

Figure 2 shows the per-class precision, recall, and F1 for scGPT pretrained. The model performs strongly on well-separated cell types: B cells (F1=0.990), CD14+ Monocytes (F1=0.944), and CD4 T cells (F1=0.934). The most challenging class is CD8 T cells (F1=0.703), where the model confuses CD8 T cells with CD4 T cells—a known biological challenge, as these populations share many marker genes and differ primarily in the expression of *CD4* and *CD8A/B*, which may not be among the top HVGs.

Importantly, the CD8 T cell challenge is not unique to scGPT: all methods in our benchmark struggle with this class to varying degrees. This suggests the difficulty is biological (marker gene overlap) rather than a model-specific failure. Megakaryocytes (F1=1.000) achieve perfect classification despite having only 2 test cells, likely because their transcriptional profile is highly distinctive (high *PPBP*, *PF4* expression).

4.3 Multi-Metric Profile Analysis

Figure 3 presents a radar chart comparing all seven methods across all seven metrics simultaneously. Several patterns emerge:

- **Deep MLP and Seurat-KNN** show the largest radar area, indicating consistently strong performance across all metrics.
- **scGPT pretrained** shows a balanced but uniformly lower profile, with particularly lower ARI (0.816) and NMI (0.806) compared to the top methods.
- **PCA+KNN** shows an interesting asymmetry: high Macro Precision (0.941) but lower Macro Recall (0.884), suggesting it is conservative in its predictions.
- **TOSICA** achieves a well-rounded profile despite having 300× fewer parameters than scGPT, suggesting that pathway-informed inductive biases can substitute for large-scale pretraining on immune cell data.

4.4 The Data-Regime Hypothesis

Our results are consistent with the data-regime hypothesis: the advantage of large-scale pretraining is most pronounced on large, complex, or heterogeneous datasets, and diminishes on small, well-separated

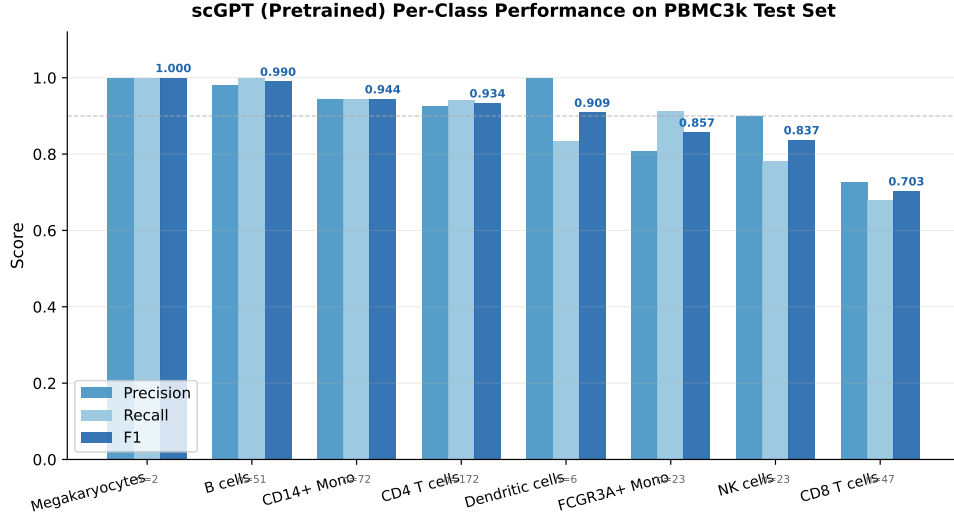


Figure 2: Per-class precision, recall, and F1 for scGPT (pretrained) on the PBMC3k test set. Cell types are sorted by F1 score (descending). Sample sizes (n) are shown below each group. CD8 T cells (F1=0.703) represent the most challenging class, reflecting the known difficulty of distinguishing CD4 and CD8 T cell populations.

benchmarks.

This is not a criticism of scGPT. The original scGPT paper [Cui et al., 2024] demonstrates clear advantages on more challenging datasets: hPancreas (~16k cells, 14 cell types), Multiple Sclerosis (~25k cells), and Myeloid (~18k cells). On these datasets, the pretrained model’s ability to leverage representations learned from 33 million cells provides a meaningful advantage. On PBMC3k—a small, clean, well-separated dataset that has been used as a tutorial example for years—the task is simply not challenging enough to reveal the pretrained model’s advantages.

This finding has a direct practical implication: *practitioners should benchmark lightweight alternatives before deploying a 51M-parameter foundation model on small datasets.* A 3-layer MLP trained in 14 seconds on CPU achieves higher Macro F1 than scGPT pretrained on PBMC3k. For rapid prototyping, exploratory analysis, or resource- constrained environments, classical and lightweight methods offer a better accuracy-to-computational-cost ratio.

4.5 Computational Cost

Table 2 summarizes approximate training times on CPU. Classical methods complete in under 1 second; the Deep MLP trains in 14 seconds; TOSICA requires approximately 12 minutes due to its attention mechanism over 300 pathways; scGPT inference (frozen backbone) requires approximately 2–5 minutes per epoch for feature extraction, with 50 epochs of classifier training.

Table 2: Approximate training/inference time on CPU (Intel Xeon, single core).

Method	Approx. Time	Parameters
PCA + LogReg	<1 s	—
PCA + KNN	<1 s	—
Seurat (PCA+KNN)	<1 s	—
Deep MLP (scratch)	~14 s	0.54M
scVI/scANVI	~15 s	0.28M
scGPT (pretrained)	~5 min	51.3M (frozen) + 0.13M (head)
TOSICA	~12 min	0.17M

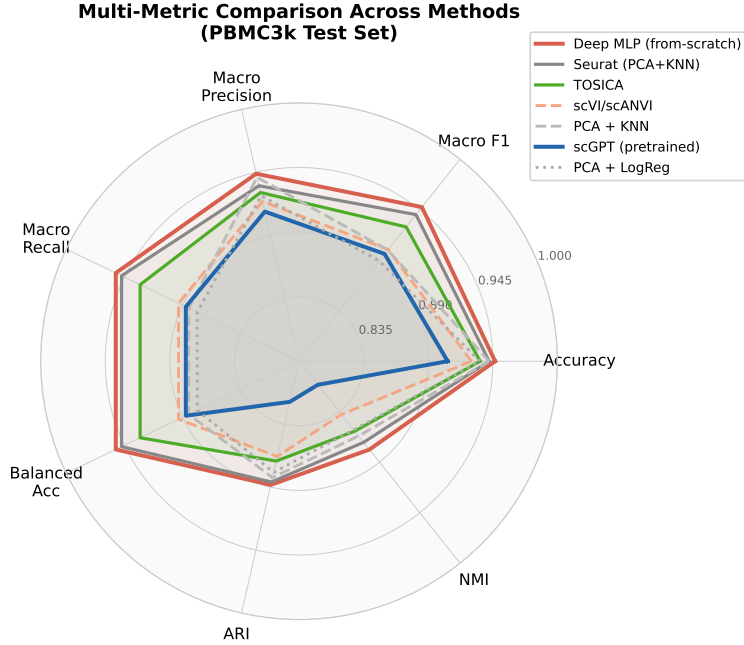


Figure 3: Radar chart comparing all seven methods across seven evaluation metrics on the PBMC3k test set. The radial axis is scaled between 0.78 and 1.00. scGPT (pretrained, blue) shows a balanced profile but consistently falls below the top methods. The Deep MLP (red) and Seurat-KNN (gray) show the largest overall area.

5 Discussion

When to use scGPT. Our results suggest that scGPT’s pretrained representations are most valuable in scenarios where (1) the dataset is large ($>10k$ cells), (2) cell types are rare or poorly characterized, (3) multi-dataset transfer is required (e.g., annotating a new dataset using a reference atlas), or (4) zero-shot annotation is needed without any labeled training data. In these regimes, the 33-million-cell pretraining corpus provides representations that cannot be learned from the task-specific data alone.

When classical methods suffice. For small datasets ($<5k$ cells) with well-characterized cell types and sufficient labeled training data, classical methods (Seurat-KNN, Logistic Regression) and lightweight deep learning models (MLP, TOSICA) offer competitive or superior performance at a fraction of the computational cost. This is consistent with findings in tabular deep learning, where simple models frequently outperform complex architectures on small datasets [Grinsztajn et al., 2022].

The role of pathway-informed inductive biases. TOSICA’s competitive performance (Macro F1=0.926, 3rd place) despite having $300\times$ fewer parameters than scGPT is noteworthy. The pathway-masked attention mechanism incorporates prior biological knowledge (immune pathway structure) directly into the model architecture, providing a form of domain-specific regularization that is particularly effective on immune cell data. This suggests that domain-specific architectural priors can be a powerful alternative to large-scale pretraining when the domain is well-characterized.

Limitations. This study has several important limitations:

1. **Single dataset.** We benchmark on PBMC3k only. Results may not generalize to other datasets, particularly those with more cell types, greater technical noise, or rarer populations. A comprehensive benchmark would include hPancreas, Multiple Sclerosis, and Myeloid datasets from the original scGPT paper.

2. **Frozen backbone only.** We evaluate scGPT with a frozen pretrained backbone. Fine-tuning the full model (unfrozen) may yield substantially higher performance, particularly on this dataset where the pretrained representations may not be optimally aligned with the PBMC3k cell type boundaries.
3. **Reimplementations.** scVI/scANVI and TOSICA are reimplementations rather than the official packages, due to dependency conflicts in our environment. While we preserved the core architectures, minor implementation differences may affect results.
4. **MLP as “from-scratch” ablation.** Due to CPU constraints, the “from-scratch” baseline uses an MLP rather than a Transformer architecture. A Transformer trained from random initialization would be a more direct ablation of scGPT’s pretraining, but was computationally infeasible on CPU (>90 minutes per run).
5. **No hyperparameter tuning.** All methods use default or lightly tuned hyperparameters. Systematic hyperparameter optimization might change the relative rankings, particularly for methods sensitive to learning rate and regularization.

Future work. Several directions would strengthen these findings: (1) extending the benchmark to the datasets used in the original scGPT paper to reproduce and contextualize its findings; (2) evaluating fine-tuned (unfrozen) scGPT to separate the effects of architecture from pretraining; (3) including Geneformer and scBERT for a more comprehensive foundation model comparison; (4) systematically varying dataset size to identify the crossover point where pretraining begins to provide a consistent advantage.

6 Conclusion

We present a systematic benchmark of seven cell type annotation methods on the PBMC3k dataset, finding that on this small, well-separated benchmark, classical methods and a lightweight Deep MLP match or exceed scGPT pretrained in Macro F1 (0.948 vs. 0.897), while all methods cluster within a narrow 4-percentage-point accuracy band. These results support a data-regime hypothesis: the advantage of large-scale pretraining is most pronounced on large, complex datasets, and diminishes on small, well-characterized benchmarks. scGPT’s frozen backbone still achieves 90.7% accuracy with zero task-specific architecture design, demonstrating that pretraining provides a strong initialization even when it does not yield the highest absolute performance.

Our primary recommendation for practitioners is to benchmark lightweight alternatives before deploying a 51M-parameter foundation model on small datasets. The computational savings are substantial (seconds vs. minutes), and the performance difference on easy benchmarks is small. For large, complex, or multi-dataset scenarios, scGPT and other foundation models remain compelling choices.

All code, trained models, and evaluation scripts are available at <https://github.com/junior1p/scgpt-pbmc68k-finetune>

References

- T. Abdelaal, L. Michielsen, D. Cats, D. Hoogduin, H. Mei, M. J. T. Reinders, and A. Mahfouz. A comparison of automatic cell identification methods for single-cell RNA sequencing data. *Genome Biology*, 20:194, 2019. doi: 10.1186/s13059-019-1795-z.
- R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.
- J. Chen, H. Xu, W. Tao, Z. Chen, Y. Zhao, and J.-D. J. Han. Interpretable machine learning for single-cell transcriptomics: a pathway-based approach. *Nature Communications*, 14:964, 2023. doi: 10.1038/s41467-023-36559-0.

- H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, and B. Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21:1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- L. Grinsztajn, E. Oyallon, and G. Varoquaux. Why tree-based models still outperform deep learning on tabular data. In *Advances in Neural Information Processing Systems*, volume 35, pages 507–520, 2022.
- M. Hao, J. Gong, X. Zeng, C. Liu, Y. Guo, X. Cheng, T. Wang, J. Ma, X. Zhang, and L. Song. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21:1481–1491, 2024. doi: 10.1038/s41592-024-02305-7.
- Y. Hao, S. Hao, E. Andersen-Nissen, W. M. Mauck, S. Zheng, A. Butler, M. J. Lee, A. J. Wilk, C. Darby, M. Zager, et al. Integrated analysis of multimodal single-cell data. *Cell*, 184:3573–3587, 2021. doi: 10.1016/j.cell.2021.04.048.
- L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi: 10.1007/BF01908075.
- V. Y. Kiselev, A. Yiu, and M. Hemberg. scmap: projection of single-cell RNA-seq data across data sets. *Nature Methods*, 15:359–362, 2018. doi: 10.1038/nmeth.4644.
- R. Lopez, J. Regier, M. B. Cole, M. I. Jordan, and N. Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15:1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.
- M. D. Luecken, M. Büttner, K. Chaichoompu, A. Danese, M. Interlandi, M. F. Mueller, D. C. Strobl, L. Zappia, M. Dugas, M. Colomé-Tatché, et al. Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19:41–50, 2022. doi: 10.1038/s41592-021-01336-8.
- A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32, 2019.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- A. Strehl and J. Ghosh. Cluster ensembles — a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617, 2002.
- T. Stuart, A. Butler, P. Hoffman, C. Hafemeister, E. Papalexi, W. M. Mauck, Y. Hao, M. Stoeckius, P. Smibert, and R. Satija. Comprehensive integration of single-cell data. *Cell*, 177:1888–1902, 2019. doi: 10.1016/j.cell.2019.05.031.
- C. V. Theodoris, L. Xiao, A. Chopra, M. D. Chaffin, Z. R. Al Sayed, M. C. Hill, H. Mantineo, E. M. Brydon, Z. Zeng, X. S. Liu, et al. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023. doi: 10.1038/s41586-023-06139-9.
- F. A. Wolf, P. Angerer, and F. J. Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15, 2018. doi: 10.1186/s13059-018-1406-4.
- C. Xu, R. Lopez, E. Mehlman, J. Regier, M. I. Jordan, and N. Yosef. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular Systems Biology*, 17:e9620, 2021. doi: 10.15252/msb.20209620.
- F. Yang, W. Wang, F. Wang, Y. Fang, D. Tang, J. Huang, H. Lu, and J. Yao. scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data. *Nature Machine Intelligence*, 4: 852–866, 2022. doi: 10.1038/s42256-022-00534-z.

G. X. Y. Zheng, J. M. Terry, P. Belgrader, P. Ryvkin, Z. W. Bent, R. Wilson, S. B. Ziraldo, T. D. Wheeler, G. P. McDermott, J. Zhu, et al. Massively parallel digital transcriptional profiling of single cells. *Nature Communications*, 8:14049, 2017. doi: 10.1038/ncomms14049.